Kingdom of Saudi Arabia
Ministry of High Education
Immam Mohammed Bin Saud Islamic University
College of Computer Science

Final Exam

Course Title and Code: *CS 215 - Design and Analysis of Algorithms*
Instructors: ***Dr. Emdad Khan, Dr. Yassine Daadaa***
Date: *Saturday 18/05/2013*
Duration: *10:00 – 12:00*
Student Name:

Student ID:

Total Mark
40

1. This is a closed book exam. Do not use any notes or books!
2. Show your work. Partial credit will be given. Grading will be based on correctness, clarity and neatness.
3. We suggest you to read the whole exam before beginning to work any problem.
4. There are 5 questions worth a total of 40 points, and 1 bonus question worth 6 points, on 11 pages (including this one)

Question 1: (10 points)

Consider the following instance of the {0,1} knapsack problem.

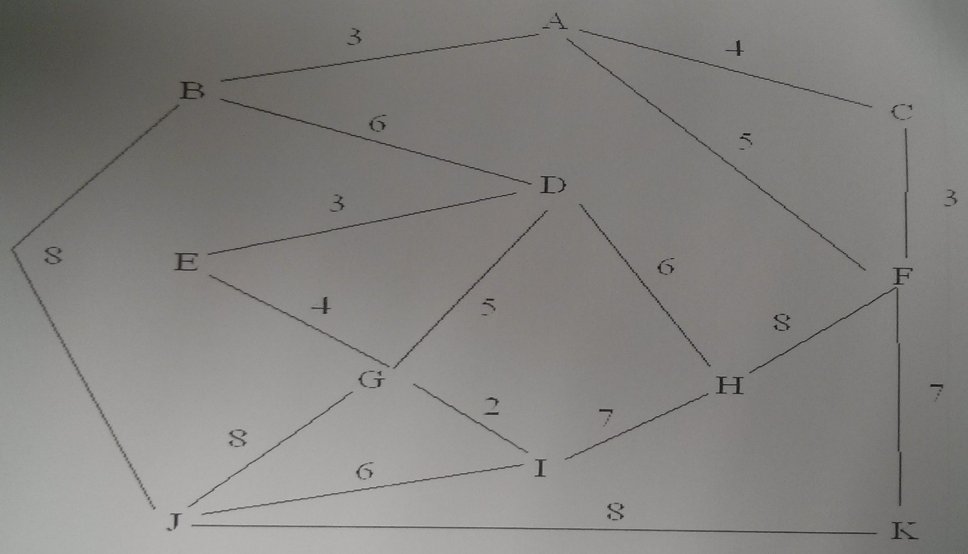
Item	a	b	c	d	e
Benefit	15	12	9	16	17
Weight	2	5	3	4	6

The maximum allowable total weight in the knapsack is Wmax = 12.

Find an optimal solution for the {0,1) Knapsack Problem given the weights and benefits above. Be sure to state both the value of the maximum benefit that you obtain as well as the item(s) that you need to obtain this benefit. Show some reasoning for your work.

Question 2: (10 points)

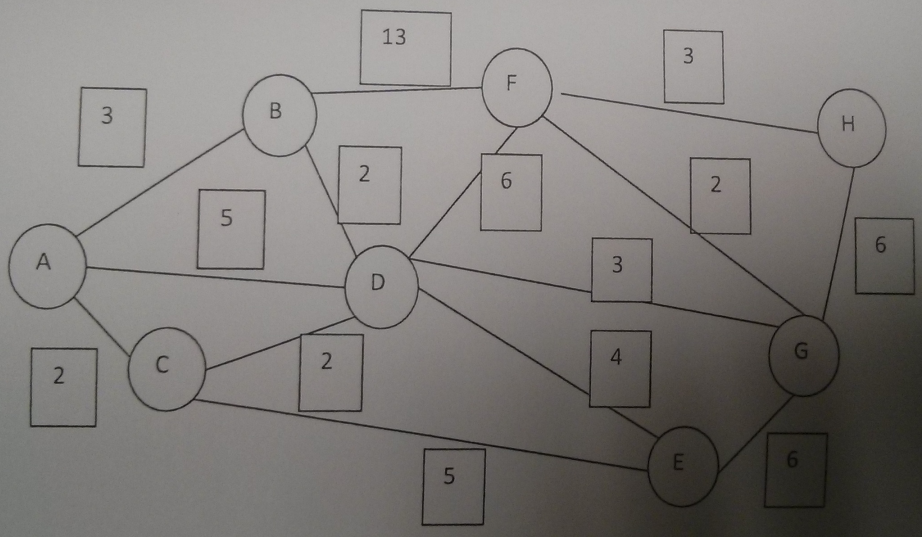
Consider the following graph:



Using Dijkstra's algorithm, determine the shortest path from node A to I. Show the steps, your tables and the resulting path.

Question 3: (10 points)

Find The Minimum Spanning Tree of the following graph using Kruskal's algorithm. Show the steps and the resulting MST.



Question 4: (10 points)

(a) Explain what are P, NP and NP-Complete problems. What is meant by "is P = NP" ?

(b) Suppose that A and B , and that A $\leq_p$ B. Circle one answer for each statement below.

If $A \in P$ then $B \in P$.

True **False**

If $B \in P$ then $A \in P$.

True **False**

If A is NP-hard then B is NP-hard.

True **False**

If B is NP-hard then A is NP-hard.

True **False**

Question 5 (bonus Question): (6 points - 2:4)

(a) Let G = (V, E) be a weighted, directed graph with exactly one negative weight edge and no negative-weight cycles. Give an algorithm to find the shortest distance from s to all other vertices in V that has the same running time as Dijkstra.

(b) A subset-sum problem is defined as follows:
A set S of positive integer and another positive integer t. Find a subset S' of integers in S such that their sum is t. For example, if S = {1,2,3,14,49,98} and t = 101, then the subset S' is {1,2,98}.

Assume that A subset-sum problem is NP-Complete. Show that the Knapsack problem is NP-Complete by reducing the subset-sum problem to Knapsack problem. [Hint: see that both problems are very similar].